

# K8s-07

## 1: Execute all yaml files shown in video.

- Create first. Yaml file and verify it

```
root@Master:~# vi first.yaml
root@Master:~# kubectl apply -f first.yaml
pod/firstpod created
root@Master:~# kubectl get pod firstpod -o wide
NAME      READY   STATUS    RESTARTS   AGE   IP            NODE      NOMINATED NODE   READINESS GATES
firstpod   1/1     Running   0           7s    192.168.142.193  worknode1  <none>           <none>
root@Master:~# cat first.yaml
apiVersion: v1
kind: Pod
metadata:
  name: firstpod
spec:
  nodeSelector:
    env: dev

  tolerations:
  - key: "special"
    operator: "Equal"
    value: "true"
    effect: "NoSchedule"

  containers:
  - name: firstcontainer
    image: nginx
    imagePullPolicy: IfNotPresent
```

- Same with in first .yaml file changed to second and check work on worknode2

```
apiVersion: v1
kind: Pod
metadata:
  name: secondpod
spec:
  nodeSelector:
    env: test

  containers:
  - name: secondcontainer
    image: nginx
    imagePullPolicy: IfNotPresent
```

```
root@Master:~# vi first.yaml
root@Master:~# kubectl apply -f first.yaml
pod/secondpod created
root@Master:~# kubectl get pod secondpod -o wide
NAME      READY   STATUS    RESTARTS   AGE   IP            NODE      NOMINATED NODE   READINESS GATES
secondpod  1/1     Running   0           9s    192.168.140.194  worknode2  <none>           <none>
```

```
apiVersion: v1
kind: Pod
metadata:
  name: firstpod
spec:
  containers:
  - name: firstcontainer
    image: nginx
    imagePullPolicy: Never
  tolerations:
  - key: "special"
    operator: "Equal"
    value: "true"
    effect: "NoSchedule"
```

```

root@Master:~# vi first.yaml
root@Master:~# kubectl apply -f first.yaml
pod/firstpod created

```

**2: Taint a Node and Schedule a Tolerant Pod** Taint a node with special=true: No Schedule. Create a pod with a toleration that matches the taint, allowing it to be scheduled on the tainted node.

- Taint worknode1 and verify

```

root@Master:~/k8s-07# kubectl taint nodes worknode1 special=true:NoSchedule-
node/worknode1 untainted
root@Master:~/k8s-07# kubectl describe node worknode1 | grep Taints
Taints:                special=true:PreferNoSchedule

```

- Label the node and verify

```

root@Master:~/k8s-07# kubectl label node worknode1 env-
node/worknode1 unlabeled
root@Master:~/k8s-07# kubectl get nodes --show-labels | grep worknode1
worknode1    Ready    <none>          10d    v1.31.14    beta.kubernetes.io/arch=amd64,beta.kuber
,kubernetes.io/os=linux

```

- Add the taint again and verify

```

root@Master:~/k8s-07# kubectl taint nodes worknode1 special=true:NoSchedule
node/worknode1 tainted
root@Master:~/k8s-07# kubectl describe node worknode1 | grep Taints
Taints:                special=true:NoSchedule
root@Master:~/k8s-07# kubectl label node worknode1 env=dev
node/worknode1 labeled

```

Label the node again and verify

```

root@Master:~/k8s-07# kubectl get nodes --show-labels | grep worknode1
worknode1    Ready    <none>          10d    v1.31.14    beta.kubernetes.io/arch=amd64,beta.kuber
orknode1,kubernetes.io/os=linux

```

- Create yaml and written yaml script

```

root@Master:~/k8s-07# vi task2.yaml
root@Master:~/k8s-07# kubectl apply -f task2.yaml
pod/task2-pod created

```

```

apiVersion: v1
kind: Pod
metadata:
  name: task2-pod
spec:
  nodeSelector:
    env: dev
  tolerations:
  - key: "special"
    operator: Equal
    value: "true"
    effect: NoSchedule
  containers:
  - name: nginx
    image: nginx
    imagePullPolicy: IfNotPresent

```

- Apply and verify it

```
root@Master:~/k8s-07# kubectl get pod task2-pod -o wide
NAME      READY   STATUS    RESTARTS   AGE   IP            NODE      NOMINATED NODE   READINESS GATES
task2-pod 1/1     Running   0           20s   192.168.142.195  worknode1  <none>           <none>
```

**3: Use Node Selector to Schedule a Pod on a Specific Node Label a node with env=dev. Create a pod with a node Selector that schedules it only on the node labeled env=dev.**

- Check Label worknode2 and verify it

```
root@Master:~/k8s-07# kubectl label node worknode2 env=test --overwrite
node/worknode2 not labeled
root@Master:~/k8s-07# kubectl get nodes --show-labels | grep worknode2
worknode2   Ready    <none>           10d   v1.31.14   beta.kubernetes.io/arch=armv8-t
worknode2, kubernetes.io/os=linux
```

- Create task3.yaml

```
root@Master:~/k8s-07#
root@Master:~/k8s-07# vi task3.yaml
```

```
apiVersion: v1
kind: Pod
metadata:
  name: task3-pod
spec:
  nodeSelector:
    env: test

  containers:
  - name: nginx
    image: nginx
    imagePullPolicy: IfNotPresent
```

- Apply and verify it

```
root@Master:~/k8s-07# kubectl get pod task3-pod -o wide
NAME      READY   STATUS    RESTARTS   AGE   IP            NODE      NOMINATED NODE   READINESS GATES
task3-pod 1/1     Running   0           37s   192.168.140.195  worknode2  <none>           <none>
```

```
root@Master:~/k8s-07# kubectl describe node worknode2 | grep Taints
Taints:                special=true:PreferNoSchedule
root@Master:~/k8s-07#
```

**4: Use Node Affinity with Soft Scheduling Label a node with env=test. Create a pod with Preferred During Scheduling Ignored During Execution node affinity, preferring to schedule it on a node labeled env=test. Remove the label and verify the pod continues to run.**

- Create tas4.yaml file and written yaml script

```
root@Master:~/k8s-07# vi task4.yaml
root@Master:~/k8s-07#
```

```

apiVersion: v1
kind: Pod
metadata:
  name: task4-pod
spec:
  affinity:
    nodeAffinity:
      preferredDuringSchedulingIgnoredDuringExecution:
        - weight: 100
          preference:
            matchExpressions:
              - key: env
                operator: In
                values:
                  - test
  containers:
    - name: nginx
      image: nginx
      imagePullPolicy: IfNotPresent

```

- Apply yaml file and verify it status

```

root@Master:~/k8s-07# kubectl apply -f task4.yaml
pod/task4-pod created
root@Master:~/k8s-07# kubectl get pod task4-pod -o wide

```

| NAME      | READY | STATUS  | RESTARTS | AGE | IP              | NODE   | NOMINATED NODE | READINESS GATES |
|-----------|-------|---------|----------|-----|-----------------|--------|----------------|-----------------|
| task4-pod | 1/1   | Running | 0        | 6s  | 192.168.219.103 | master | <none>         | <none>          |

```

root@Master:~/k8s-07#

```

- Remove the label and Verify the Pod

```

root@Master:~/k8s-07# kubectl label node worknode2 env-
node/worknode2 unlabeled
root@Master:~/k8s-07# kubectl get nodes --show-labels | grep worknode2

```

| NAME      | READY | STATUS | RESTARTS | AGE      | IP                                                                               | NODE | NOMINATED NODE | READINESS GATES |
|-----------|-------|--------|----------|----------|----------------------------------------------------------------------------------|------|----------------|-----------------|
| worknode2 | Ready | <none> | 10d      | v1.31.14 | beta.kubernetes.io/arch=amd64,beta.kubernetes.io/os=linux,kubernetes.io/os=linux |      |                |                 |

```

root@Master:~/k8s-07# kubectl get pod task4-pod -o wide

```

| NAME      | READY | STATUS  | RESTARTS | AGE | IP              | NODE   | NOMINATED NODE | READINESS GATES |
|-----------|-------|---------|----------|-----|-----------------|--------|----------------|-----------------|
| task4-pod | 1/1   | Running | 0        | 57s | 192.168.219.103 | master | <none>         | <none>          |

```

root@Master:~/k8s-07#

```

- Delete the existing pod env=test label and verify it still running

```

root@Master:~/k8s-07# kubectl label node worknode2 env=test
node/worknode2 labeled
root@Master:~/k8s-07# kubectl delete pod task4-pod
pod "task4-pod" deleted
root@Master:~/k8s-07# kubectl apply -f task4.yaml
pod/task4-pod created
root@Master:~/k8s-07# kubectl get pod task4-pod -o wide

```

| NAME      | READY | STATUS  | RESTARTS | AGE | IP             | NODE   | NOMINATED NODE | READINESS GATES |
|-----------|-------|---------|----------|-----|----------------|--------|----------------|-----------------|
| task4-pod | 1/1   | Running | 0        | 8s  | 192.168.219.80 | master | <none>         | <none>          |

```

root@Master:~/k8s-07# kubectl label node worknode2 env-
node/worknode2 unlabeled

```

**5: Implement Node Affinity with Hard Scheduling Create a pod with Required During Scheduling Ignored During Execution node affinity, ensuring it will only be scheduled on a node labeled env=prod. Verify the pod cannot be scheduled if no node has the env=prod label.**

- Make sure no node has env=prod

```

root@Master:~/k8s-07# kubectl get nodes --show-labels

```

| NAME      | STATUS | ROLES         | AGE | VERSION  | LABELS                                                                                                                                                                          |
|-----------|--------|---------------|-----|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| master    | Ready  | control-plane | 10d | v1.31.14 | beta.kubernetes.io/arch=amd64,beta.kubernetes.io/os=linux,kubernetes.io/os=linux,node-role.kubernetes.io/control-plane=,node.kubernetes.io/exclude-from-external-load-balancing |
| worknode1 | Ready  | <none>        | 10d | v1.31.14 | beta.kubernetes.io/arch=amd64,beta.kubernetes.io/os=linux,kubernetes.io/os=linux                                                                                                |
| worknode2 | Ready  | <none>        | 10d | v1.31.14 | beta.kubernetes.io/arch=amd64,beta.kubernetes.io/os=linux,kubernetes.io/os=linux                                                                                                |

- Create yaml file and written yaml file

```
root@Master:~/k8s-07# vi task5.yaml
root@Master:~/k8s-07#
```

```
apiVersion: v1
kind: Pod
metadata:
  name: task5-pod
spec:
  affinity:
    nodeAffinity:
      requiredDuringSchedulingIgnoredDuringExecution:
        nodeSelectorTerms:
          - matchExpressions:
              - key: env
                operator: In
                values:
                  - prod
  containers:
  - name: nginx
    image: nginx
    imagePullPolicy: IfNotPresent
```

- Apply yaml file and verify it

```
root@Master:~/k8s-07# kubectl apply -f task5.yaml
pod/task5-pod created
root@Master:~/k8s-07# kubectl get pod task5-pod
NAME          READY   STATUS    RESTARTS   AGE
task5-pod     0/1     Pending   0           42s
root@Master:~/k8s-07#
```

- Describe the Pod

```
root@Master:~/k8s-07# kubectl describe pod task5-pod
Name:          task5-pod
Namespace:     default
Priority:       0
Service Account: default
Node:          <none>
Labels:        <none>
Annotations:   <none>
Status:        Pending
IP:            <none>
IPs:           <none>
```

- Check Label worknode2 with env=prod

```
root@Master:~/k8s-07# kubectl label node worknode2 env=prod
node/worknode2 labeled
root@Master:~/k8s-07# kubectl get nodes --show-labels | grep worknode2
worknode2    Ready    <none>          10d    v1.31.14    beta.kubernetes.io/arch=armv8-t
```

- Watch the Pod and Verify the node

```
root@Master:~/k8s-07# kubectl get pod task5-pod -w
NAME          READY   STATUS    RESTARTS   AGE
task5-pod     1/1     Running   0           4m2s
^Croot@Master:~/k8s-07# kubectl get pod task5-pod -o wide
NAME          READY   STATUS    RESTARTS   AGE   IP              NODE          NOMINATED NODE   READINESS GATES
task5-pod     1/1     Running   0           4m24s  192.168.140.196  worknode2     <none>           <none>
```

**6: Taint a Node and Use No Execute with Toleration Seconds** Taint a node with special=true: No Execute. Create a pod with a toleration Seconds field (e.g., 60 seconds) and observe it gets evicted after 60 seconds on the tainted node.

- Remove the old No Schedule taint from worknode1 and verify it

```
root@Master:~/k8s-07# kubectl taint nodes worknode1 special=true:NoSchedule-  
node/worknode1 untainted  
root@Master:~/k8s-07# kubectl describe node worknode1 | grep Taints  
Taints:                special=true:PreferNoSchedule
```

- Apply a No Execute taint and verify

```
root@Master:~/k8s-07# kubectl taint nodes worknode1 special=true:NoExecute  
node/worknode1 tainted  
root@Master:~/k8s-07#
```

- Create task6.yaml

```
root@Master:~/k8s-07#  
root@Master:~/k8s-07# vi task6.yaml
```

```
apiVersion: v1  
kind: Pod  
metadata:  
  name: task6-pod  
spec:  
  nodeSelector:  
    env: dev  
  
  tolerations:  
  - key: "special"  
    operator: Equal  
    value: "true"  
    effect: NoExecute  
    tolerationSeconds: 60  
  
  containers:  
  - name: nginx  
    image: nginx  
    imagePullPolicy: IfNotPresent
```

- Apply and verify it

```
root@Master:~/k8s-07# kubectl apply -f task6.yaml
pod/task6-pod created
root@Master:~/k8s-07# kubectl get pod task6-pod -o wide
```

| NAME      | READY | STATUS  | RESTARTS | AGE | IP              | NODE      | NOMINATED NODE | READINESS GATES |
|-----------|-------|---------|----------|-----|-----------------|-----------|----------------|-----------------|
| task6-pod | 1/1   | Running | 0        | 8s  | 192.168.142.196 | worknode1 | <none>         | <none>          |

```
root@Master:~/k8s-07#
```

- Watch the pod
- It Terminates and delete task-6 pod

```
root@Master:~/k8s-07# kubectl get pods
```

| NAME                                 | READY | STATUS  | RESTARTS    | AGE |
|--------------------------------------|-------|---------|-------------|-----|
| nginx-deployment-f9db776c5-7rf56     | 1/1   | Running | 1 (54m ago) | 43h |
| recreate-deployment-7c6d446b89-m86s6 | 1/1   | Running | 1 (54m ago) | 43h |
| surge-deployment-c95fcdf96-h96rx     | 1/1   | Running | 1 (54m ago) | 43h |
| task3-pod                            | 1/1   | Running | 0           | 29m |
| task4-pod                            | 1/1   | Running | 0           | 22m |
| task5-pod                            | 1/1   | Running | 0           | 14m |

```
root@Master:~/k8s-07# kubectl get pod task6-pod -w
Error from server (NotFound): pods "task6-pod" not found
root@Master:~/k8s-07#
```